IRS-assisted Atmospheric Turbulence-Controlled Cryptosystems

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Introduction

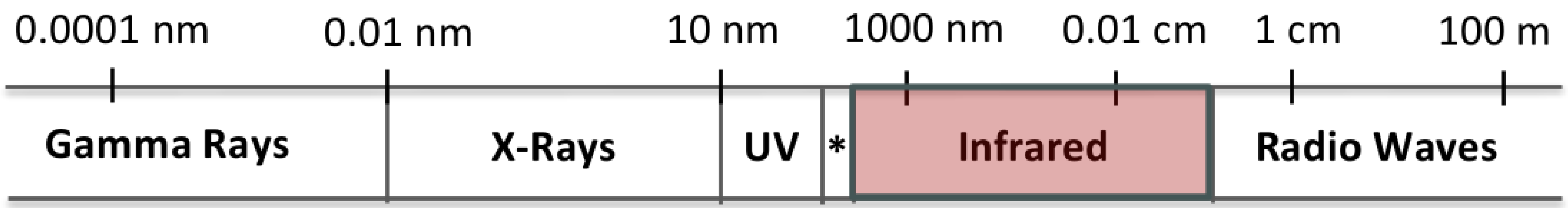
- Traditional encryption (relies on computational hardness)
  →risk that it could be broken in the future with the advancement of quantum computers

- QKD (ultimate security based on the laws of physics)
  →requires expensive equipment

- FSO (practical and high-capacity alternative)
  →cost-effectively, leveraging existing optical wireless infrastructure

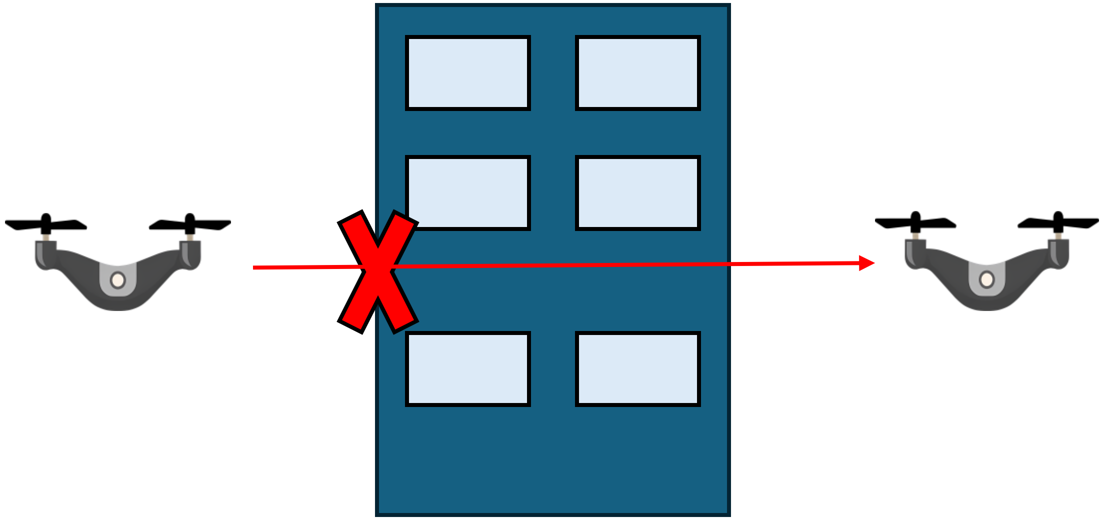
FSO : Free Space Optical

Advantages
-High data rate
-Easy to install
-License-free

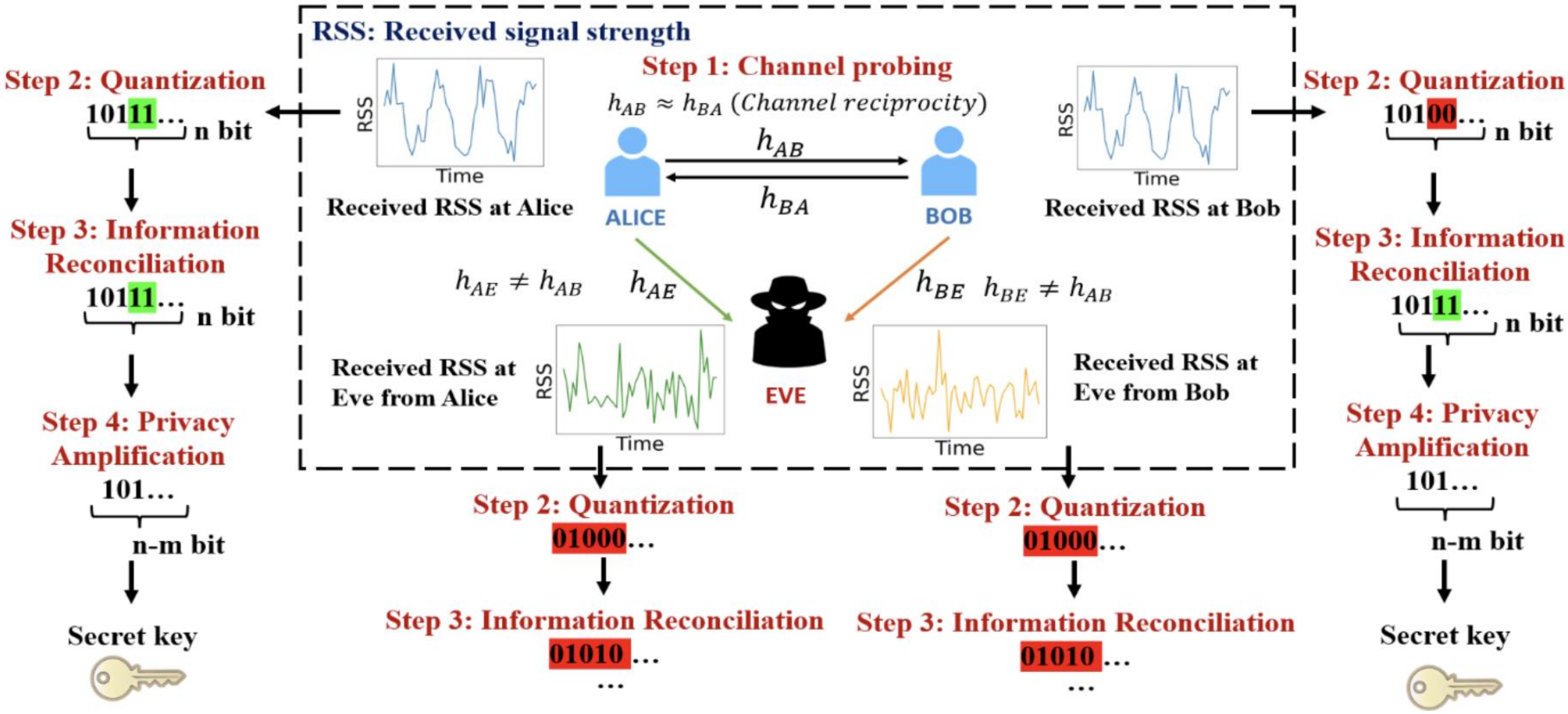


Weakness
-Need LOS(Line-of-Site)
-Affected by atmospheric turbulence

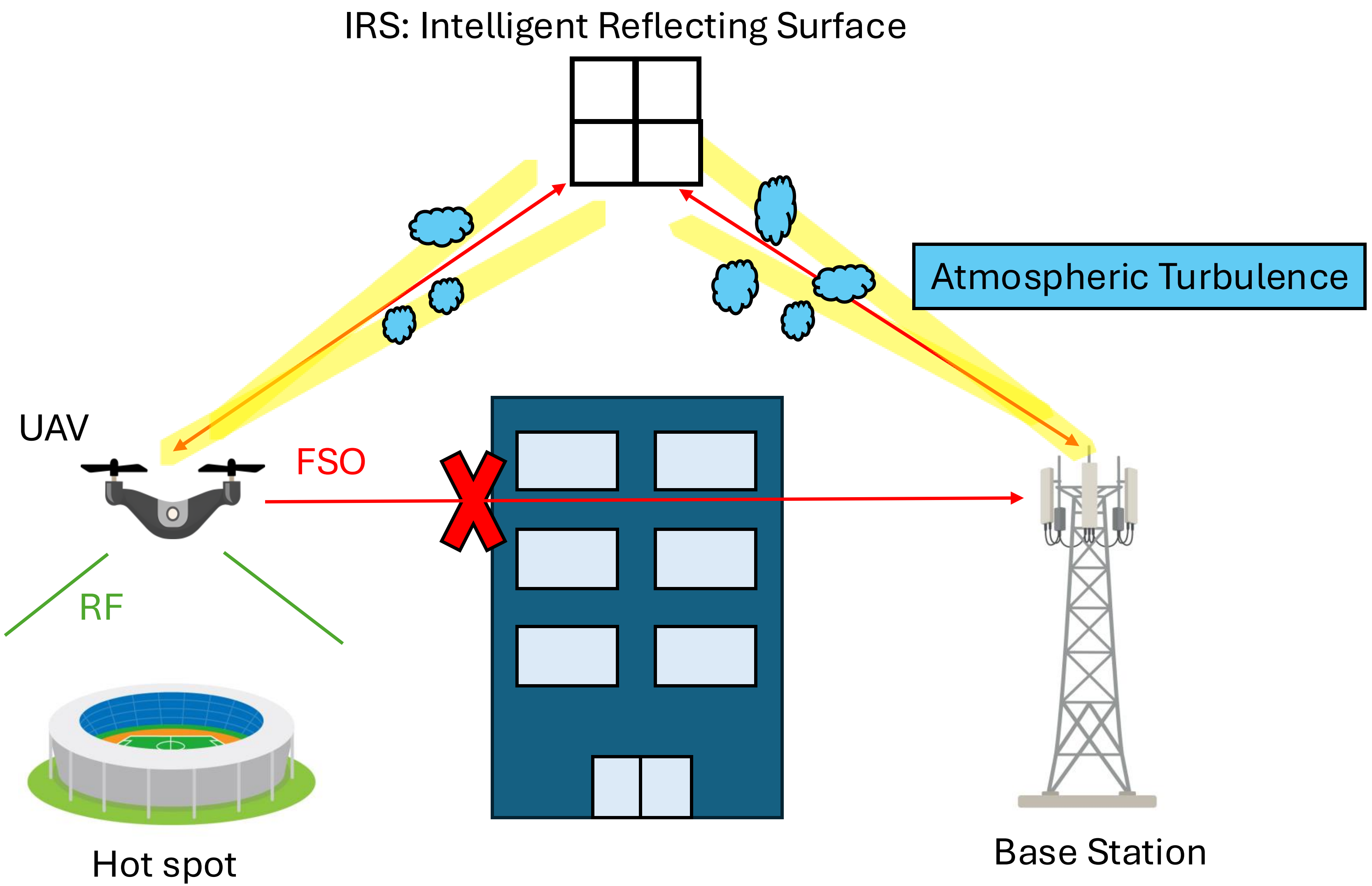
→PLS(Physical Layer Security):
using atmospheric turbulence to create secure encryption key



PLS-Key generation main steps



System Model



Atmospheric Factor

Atmospheric Attenuation :
optical energy is absorbed by air molecules or fog
→Atmospheric Transmittance Model

$$T_{\text{atmosphere}} = \exp(-\kappa \cdot L_{\text{total}})$$

standard clear weather attenuation coefficient $\kappa = 0.43\ \text{dB/km}$

Atmospheric Factor

Scintillation:
small-scale turbulent eddies act like microscopic lenses, causing the received light intensity to fluctuate randomly
→Hufnagel-Valley Boundary (HVB) model, Rytov variance

HVB model

$$C_n^2(h) = 0.00594\left(\frac{V}{27}\right)^2 (10^{-5}h)^{10} e^{-\frac{h}{1000}} + 2.7 \times 10^{-16} e^{-\frac{h}{1500}} + A e^{-\frac{h}{100}}$$

V :Grounded wind speed 21 m/s
A :Grounded-level turbulence constant 3.1 * 10^(-13) (Strong turbulence regime)

Rytov variance

$$\sigma_R^2 = 2.25 \cdot k^{7/6} \cdot L^{11/6} \int_0^1 C_n^2(h(x)) \cdot x^{5/6}\, dx$$

Atmospheric Factor

Beam Wander:
turbulent eddies larger than the beam diameter randomly displace the entire laser center, causing it to miss the target
→Beam Wander Variance formula, pointing loss factor

$$\sigma_{\text{wander}}^2 = 7.25 \cdot L_{\text{total}}^2 \cdot W_0^{-1/3} \int_0^1 C_n^2(h(x)) \cdot x^2 \, dx$$

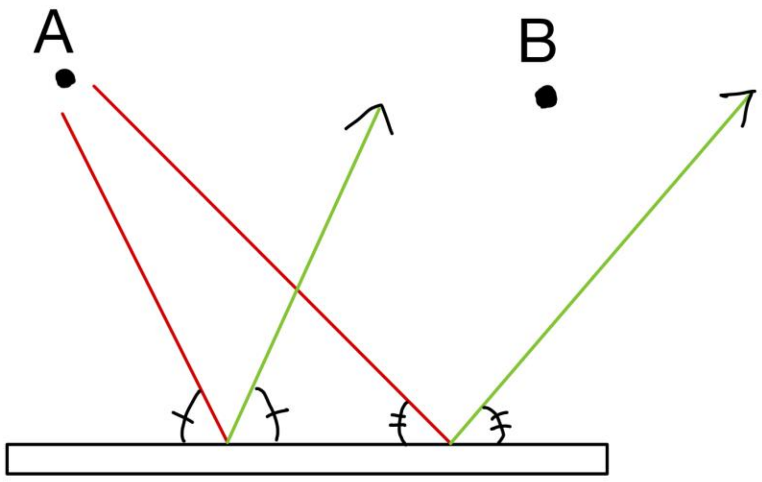
pointing loss factor

$$A_{\text{pe}} = \exp\left(-8 \frac{\sigma_{\text{wander}}^2}{\theta_{\text{div}}^2}\right)$$

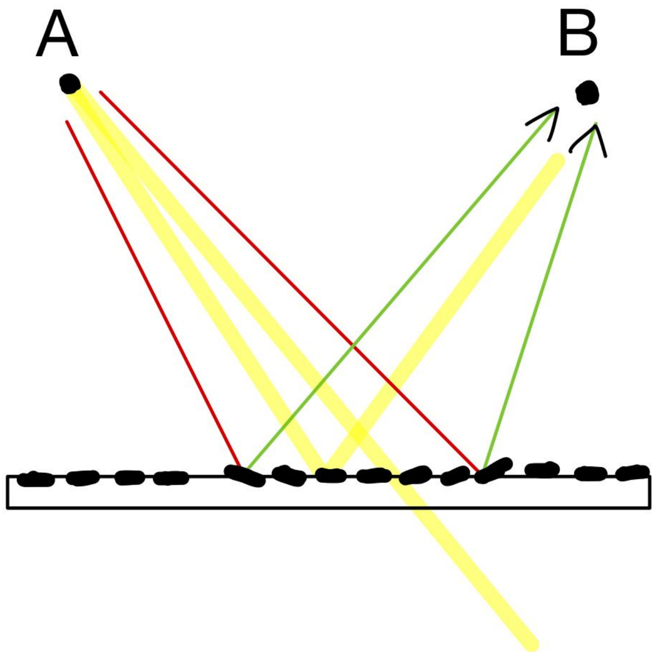
IRS Factor

Mirror-based

Single Mirror



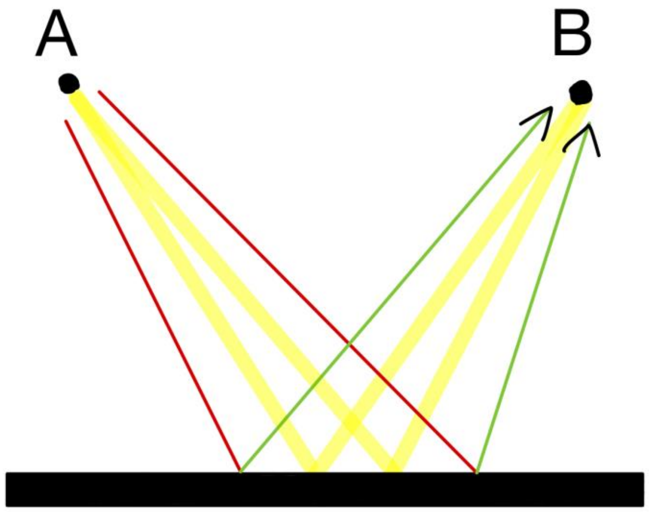
Multi Mirror



Fill-Factor loss
30% energy drops

Meta-surface-based

Reconfigurable Meta-surface



Zero physical gaps

How to calculate Secret Key Capacity

Instantaneous received SNR at Bob

$$\gamma_B = \gamma_0 \cdot G_{\text{IRS}} \cdot A_{\text{pe}} \cdot I_{\text{main}}^2$$

γ_0 : Average symbol SNR

G_{IRS} : IRS channel gain

A_{pe} : Pointing error loss factor

I_{main}^2 : Scintillation intensity squared Gamma-Gamma Distribution

$$I_{\text{main}}^2 = (X \cdot Y)^2 \quad \text{where} \quad X \sim \text{Gamma}\left(\alpha_B, \frac{1}{\alpha_B}\right), Y \sim \text{Gamma}\left(\beta_B, \frac{1}{\beta_B}\right)$$

Gamma-gamma distribution shape parameters

$$\alpha_B = \left[\exp\left(\frac{0.49 \cdot \sigma_R^2}{(1 + 1.11 \cdot \sigma_R^{12/5})^{7/6}}\right) - 1\right]^{-1} \qquad \beta_B = \left[\exp\left(\frac{0.51 \cdot \sigma_R^2}{(1 + 0.69 \cdot \sigma_R^{12/5})^{5/6}}\right) - 1\right]^{-1}$$

How to calculate Secret Key Capacity

Instantaneous received SNR at Eve

$$\gamma_E = \gamma_{0,\mathrm{E}} \cdot I_{\mathrm{eve}}^2$$

$\gamma_{0,\mathrm{E}}$: Eve's average SNR

I_{eve}^2 : Eve's scintillation intensity squared

$$I_{\mathrm{eve}}^2 = (X_E \cdot Y_E)^2 \quad \text{where} \quad X_E \sim \mathrm{Gamma}\left(\alpha_E, \frac{1}{\alpha_E}\right), Y_E \sim \mathrm{Gamma}\left(\beta_E, \frac{1}{\beta_E}\right)$$

Gamma-gamma distribution shape parameters

Fix Rytov variance to 1 $\sigma_R^2 = 1$

$$\alpha_E = 4.39$$

$$\beta_E = 2.59$$

Secret Key Capacity

Capacity of Bob and Eve

$$C_{\mathrm{B}} = \log_2 (1 + \gamma_B)$$

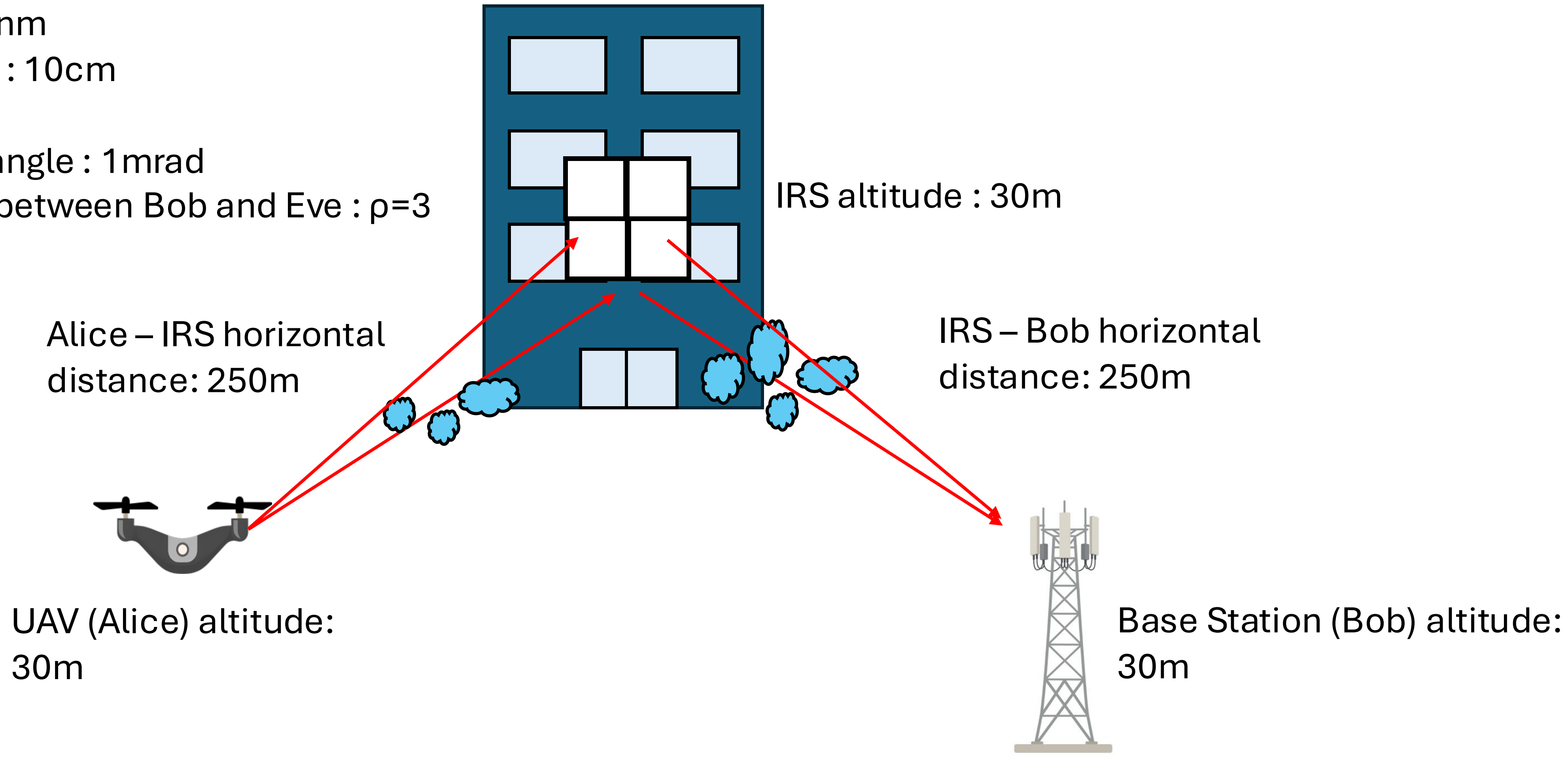
$$C_{\mathrm{E}} = \log_2 (1 + \gamma_E)$$

Secret key capacity

$$C_{\mathrm{SK}} = \mathbb{E}\left[\max\left(0,\, C_B - C_E\right)\right] \quad [\mathrm{bits/s/meas./dim}]$$

Current parameter settings

Wavelength : 1550nm
Receiver Diameter : 10cm
UAV's jitter : 0
Beam divergence angle : 1mrad
Average SNR ratio between Bob and Eve : ρ=3



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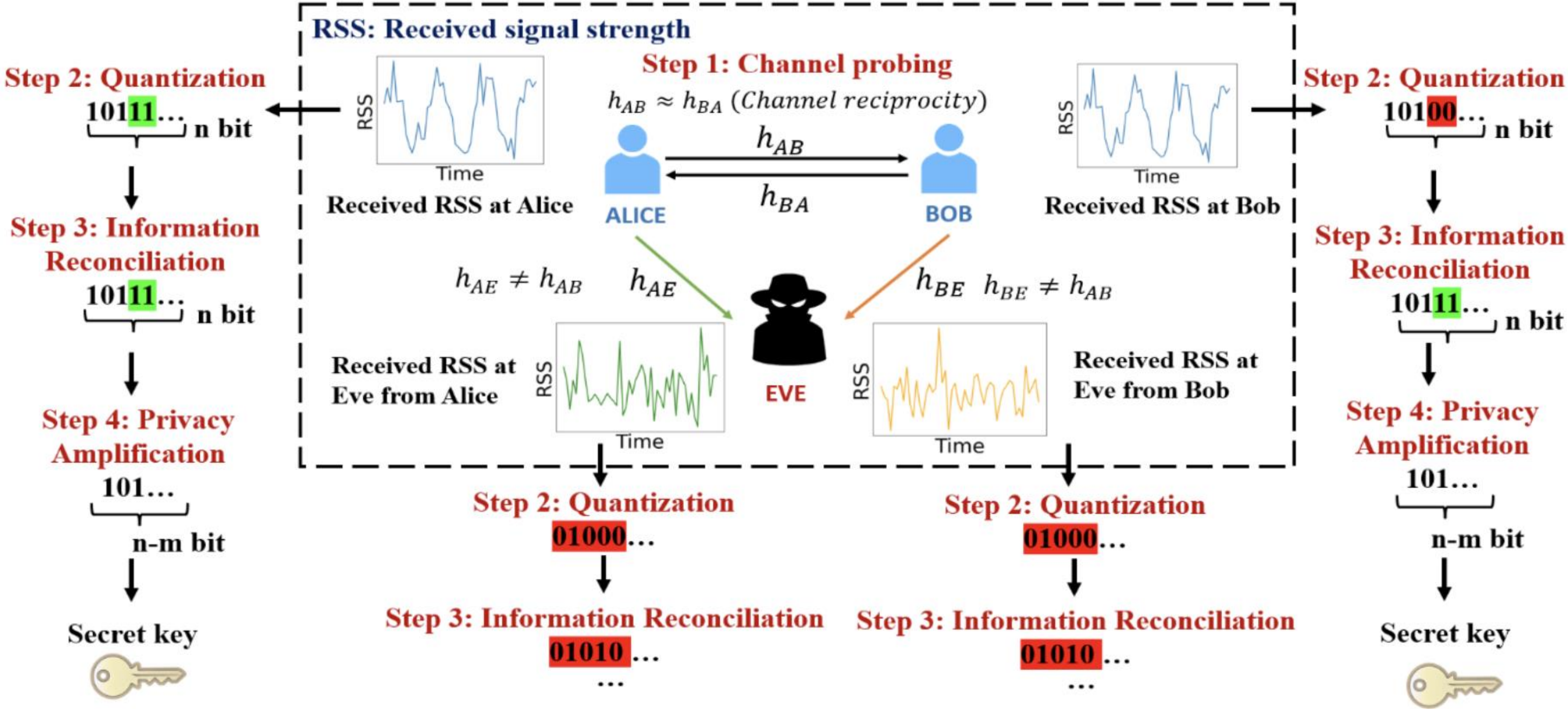

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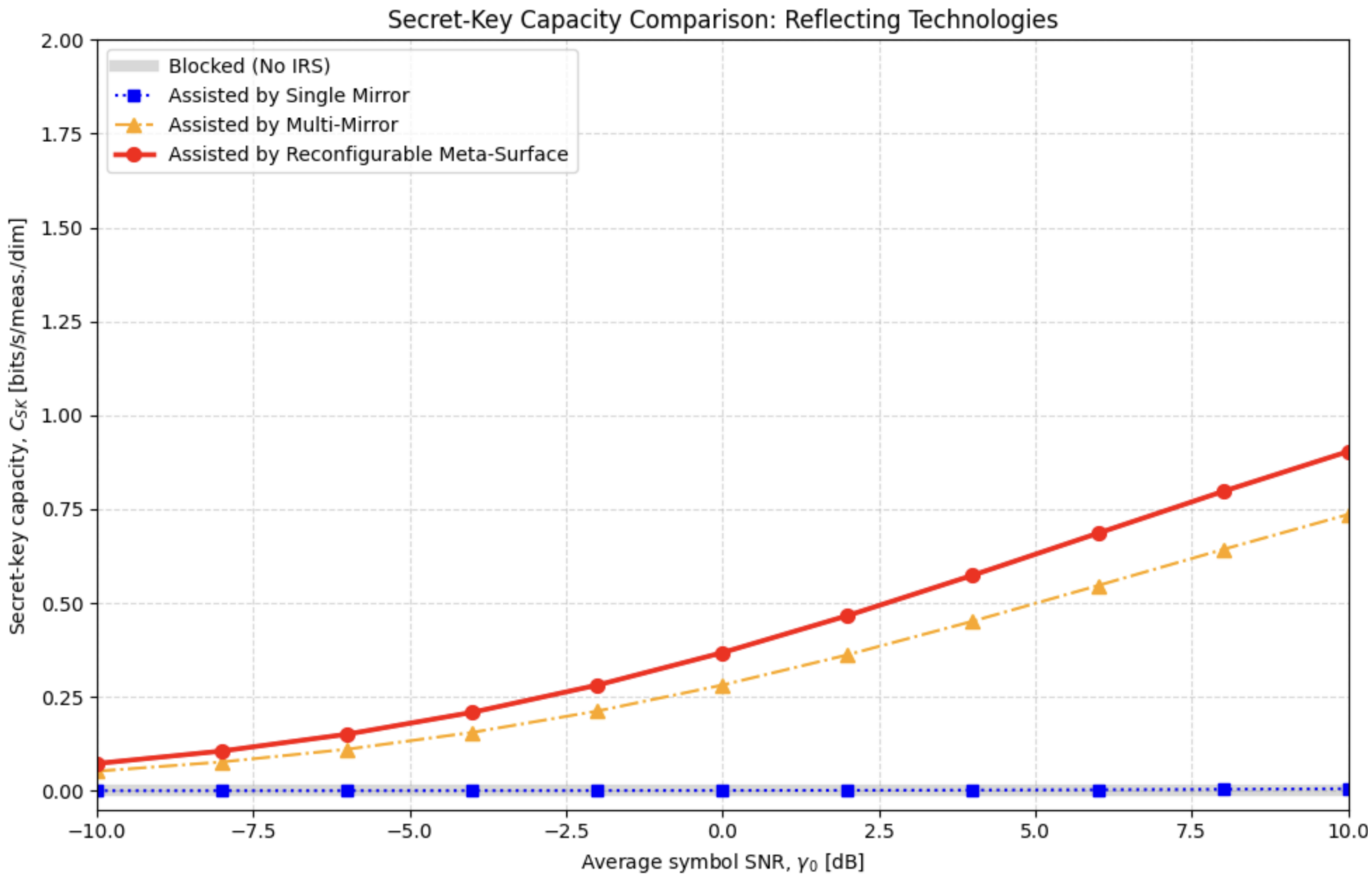
Result



Multi-Mirror and Reconfigurable Meta-Surface both capable of beam focusing. However, Multi-Mirror suffers from physical gaps between its micro-mirrors.

Next step

- Mathematical Refinement of Secrecy Metric
  -update framework to the difference between
  Ergodic capacities

- Integration of UAV Mechanical Platform Jitter

Atmospheric Turbulence Controlled Cryptosystems

Alice

Raw Key

00111101…

CSP RNG

Bob

Raw Key

00111101…

CSP RNG

Extract seed to initialized CSPRNG

Generate <u>random number sequence</u>

Alice	Bob
Gamma-Gamma Shaping (Inverse CDF)	Gamma-Gamma Shaping (Inverse CDF)
CDF-Based Quantization (0.5 threshold)	CDF-Based Quantization (0.5 threshold)

Every Tc seconds (Coherence time of atmosphere), update **α and β (parameter of Gamma-Gamma distribution)**

Secret Key

0101010101….

0101010101….

used for One-Time Pad encryption and then immediately destroyed

IRS-assisted Atmospheric Turbulence-Controlled Cryptosystems

Result